

Introduction to Probability

Part 2

Stat 3113: Lesson 7

Calculating the Probability of the Event A : $P(A)$

Three Axioms of Probability

Axiom 1: For any event A , $P(A) \geq 0$.

Axiom 2: $P(\text{sample space}) = 1$

Axiom 3: If $A_1, A_2, \dots, A_n$ is a finite collection of *mutually exclusive* (or *disjoint*) events, then

$$P(A_1 \cup A_2 \cup \dots \cup A_n) = P(A_1) + P(A_2) + \dots + P(A_n).$$

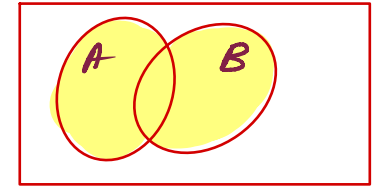
Note: For just two disjoint events, A and B , $P(A \cup B) = P(A) + P(B)$.

Probability Rules – derived from 3 Axioms

- 1. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$ *(Inclusion-Exclusion formula)*
- 2. $P(A^C) = 1 - P(A)$
- 3. $P(\emptyset) = 0$
- 4. $P(A) = P(A \cap B) + P(A \cap B^C)$
- 5. $P(A) = P(A \cup B) - P(A^C \cap B)$

Probability Rules – derived from 3 Axioms

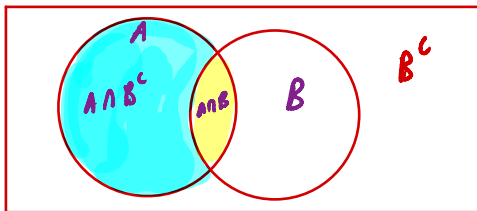
$$\begin{aligned}
 1) \quad IP(A \cup B) &= IP((A \cap B^c) \cup (A \cap B) \cup (B \cap A^c)) \\
 &= IP(A \cap B^c) + IP(A \cap B) + IP(B \cap A^c) \\
 &= [IP(A) - IP(A \cap B)] + IP(A \cap B) + [IP(B) - IP(A \cap B)] \\
 &= IP(A) + IP(B) - IP(A \cap B)
 \end{aligned}$$



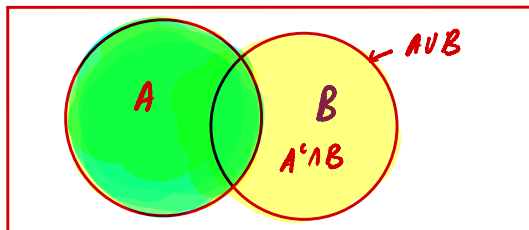
$$2) IP(S) = IP(A \cup A^c) = IP(A) + IP(A^c) = 1 \Rightarrow IP(A^c) = 1 - IP(A)$$

$$3) IP(\emptyset) = IP(S^c) = 1 - IP(S) = 0$$

4)



5)



Example 1: If A and B are disjoint $P(A) = 0.4$ and $P(B) = 0.5$.

Find:

a. $P(A \cup B) = P(A) + P(B) = 0.4 + 0.5 = 0.9$

b. $P(A^c) = 1 - P(A) = 1 - 0.4 = 0.6$

c. $P(B \cap A^c) = P(A \cup B) - P(A)$
 $= P(B) = 0.5$

Example 2: A family that owns two automobiles is selected at random. Let A_1 be the event the older car is American and A_2 be the event the newer car is American. If $P(A_1) = 0.7$, $P(A_2) = 0.5$, and $P(A_1 \cap A_2) = 0.4$, compute the following:

a. The probability at least one car is American.

$$\begin{aligned} P(A_1 \cup A_2) &= P(A_1) + P(A_2) - P(A_1 \cap A_2) \\ &= 0.7 + 0.5 - 0.4 \\ &= 0.8 \end{aligned}$$

b. The probability that older car is not American.

$$\begin{aligned} P(A_1^c) &= 1 - P(A_1) \\ &= 1 - 0.7 \\ &= 0.3 \end{aligned}$$

Example 3: A mutual fund company offers its customers several different funds: a money-market fund, three different bond funds (short, intermediate, and long-term), two stock funds (moderate and high-risk), and a balanced fund. Among customers who own shares in just one fund, the percentages of customers in the different funds are as follows:

Money-market	20%
Short bond	15%
Intermediate bond	10%
Long bond	5%
High-risk stock	18%
Moderate-risk stock	25%
Balanced	7%

↳ Must add up to 100%

A customer who owns shares in just one fund is randomly selected.

a) What is probability that the individual owns shares in the balanced fund?

$$= 7\%$$

b) What is the probability that the individual owns shares in a bond fund?

$$= 15\% + 10\% + 5\%$$

$$= 30\%$$

c) What is the probability that the individual does not own shares in a stock fund?

$$= 100\% - (18\% + 25\%)$$

$$= 57\%$$

Equally Likely Outcomes

When an experiment has N equally likely outcomes, the probability of an event A occurring is:

$$P(A) = \frac{N(A)}{N}$$

where: $N(A)$ – number of ways the event A can occur

N – number of equally likely outcomes

Example: A statistics class for engineers consists of 25 industrial, 10 mechanical, 10 electrical, and 8 civil engineering students. If a person is randomly selected by an instructor to answer a question, find the probability that the student chosen is:

a) an industrial engineering major.

$$P(\text{Industrial}) = \frac{25}{25 + 10 + 10 + 8} = \frac{25}{53} \approx 47\%$$

b) a civil engineering or an electrical engineering major.

Equally Likely Outcomes

$$\begin{aligned} P(\{Civil\} \cup \{Electrical\}) &= P(\{Civil\}) + P(\{Electrical\}) \\ &= \frac{8}{53} + \frac{10}{53} \\ &\approx 34\% \end{aligned}$$

Reading

Chapter 3 – Section 3.1

Next Topic



Conditional Probability



Chapter 3 - Section 3.2